Aghar Usman Kannanthodi

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Professional Summary

Backend Developer specializing in AI/ML systems, LLM engineering, and scalable API development. Experienced building production-grade systems with Python, Ollama, Flask, REST APIs, MSSQL/PostgreSQL, and NLP pipelines.

Education

Malnad College of Engineering — Hassan, Karnataka, India
B.E. Information Science and Engineering | CGPA: 7.81/10 (First Class with Distinction) — *Aug 2021 – May 2025*

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Machine Learning, Artificial Intelligence, Big Data Technologies, Software Engineering, Computer Networks

Technical Skills

Languages: Python (Advanced), JavaScript, Java, C, SQL
Backend & APIs: Flask, FastAPI, REST APIs, Microservices, Rate Limiting
Databases: MSSQL, PostgreSQL, MySQL, SQLite, pyodbc, psycopg2
AI/ML & LLMs: Ollama, Llama 3, Mistral, BERT, Transformers, PyTorch, TensorFlow, Scikit-learn, HuggingFace
Cloud & DevOps: GCP, Docker, Linux/Unix, Git/GitHub, CI/CD
Data & Tools: Pandas, NumPy, NLTK, spaCy, TF-IDF, FuzzyWuzzy, OpenCV, React.js

Professional Experience

AI Intern — Feb 2025 – May 2025
Team Thai — *Calicut, Kerala*

- Built a production-ready Text-to-SQL chatbot using Ollama (Llama 3/Mistral) with **SQL injection prevention** via 7-pattern regex detection and SELECT-only enforcement for secure natural language querying over an MSSQL sales database.
- Designed a dual-model pipeline with **3-attempt retry logic**, thread-safe TTL caching, and prompt engineering with 14 few-shot examples across 4 tables, achieving accurate semantic parsing with rate limiting (10 req/min).
- Deployed a Flask REST API supporting configurable **pagination (10–500 rows)**, CSV export, and concurrent request handling via thread-pool executor across MSSQL and PostgreSQL backends.
- Implemented a hybrid search engine combining SQL with **fuzzy matching** (FuzzyWuzzy token-set/partial-ratio) and result deduplication, cutting average query response time by ~**35%** via caching and batch fetching.
- Built a PostgreSQL ETL pipeline migrating a **22-column customer schema** from CSV with batch insertion (100 rows/batch), per-row retry, timestamp normalisation, and automated failed-row reporting.
- **Technologies:** Python, Flask, Ollama, Llama 3, Mistral, MSSQL, PostgreSQL, pyodbc, psycopg2, FuzzyWuzzy, Pandas, sentence-transformers

Technical Projects

AgriLeaf Pro | *CNN-based Plant Disease Detection with IoT Integration* | GitHub — 2024 – 2025

- Developed a 5-layer CNN classifying **19 plant diseases** across 4 crops with 85%+ accuracy and under 100ms inference for real-time deployment.
- Built a Flask REST API with OpenCV preprocessing, secure authentication, and automated treatment recommendations; integrated ThingSpeak IoT pipeline monitoring **6 soil parameters** with 7-day visualization.
- *Stack: TensorFlow, Flask, OpenCV, IoT, GCP*

Additional Information

Achievement: *State-Level Hackathon 2nd Place* — Led a 4-member team in a 6-hour hackathon, building a college study material platform.
Languages: English (Professional), Hindi (Professional), Malayalam (Conversational), Kannada (Conversational)